

Static Two-Phase Updated-Lagrangian MPM

Implementation Derivation Notes

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1. Scope

This note documents the formulation implemented in `src/igampm/MPM/static_twophase_ulmpm.py`.

The target is the static two-phase updated-Lagrangian MPM formulation in `docs/GeoWarp-main/ImplicitTwo` cross-checked against `docs/GeoWarp-main/mpm/mpm_utils.py` and `docs/GeoWarp-main/solvers/quasi_stat`

The only intended constitutive simplification relative to the full paper is that the solid skeleton is currently restricted to elastic laws: `linearElastic` and `neoHookean`. The kinematics, residual structure, pressure projection, PPP stabilization, and hand-written Jacobian blocks follow the paper/reference implementation path.

2. Nodal Unknowns and Particle Interpolation

At each active node I , the unknown is

$$\mathbf{q}_I = \begin{bmatrix} u_{I,x} & u_{I,y} & p_I \end{bmatrix}^T.$$

For one particle p , with standard shape N_I , averaged shape $\bar{N}_I$, and reference gradient $\nabla_X N_I$, define

$$\begin{aligned} \Delta \mathbf{u}_p &= \sum_I N_I \mathbf{u}_I, & \nabla_X \mathbf{u}_p &= \sum_I \mathbf{u}_I \otimes \nabla_X N_I, \\ p_p &= \sum_I N_I p_I, & p_{p,n} &= \sum_I N_I p_{I,n}, \\ \bar{p}_p &= \sum_I \bar{N}_I p_I, & \bar{p}_{p,n} &= \sum_I \bar{N}_I p_{I,n}. \end{aligned}$$

3. Updated-Lagrangian Kinematics

The incremental deformation gradient is

$$\Delta \mathbf{F}_p = \mathbf{I} + \nabla_X \mathbf{u}_p, \quad \mathbf{A}_p = \Delta \mathbf{F}_p^{-1}.$$

The updated total deformation gradient and Jacobian are

$$\mathbf{F}_p^{n+1} = \Delta \mathbf{F}_p \mathbf{F}_p^n, \quad J_p^{n+1} = \det(\mathbf{F}_p^{n+1}), \quad J_p^n = \det(\mathbf{F}_p^n).$$

The current particle volume is

$$v_p^{n+1} = v_p^0 J_p^{n+1}.$$

The pushed-forward gradient used in the weak form is

$$\nabla_x N_I = \mathbf{A}_p \nabla_X N_I.$$

Hence the particle pressure gradient is

$$\nabla_x p_p = \sum_I p_I \nabla_x N_I.$$

4. Two-Phase State Update

With initial porosity ϕ^0 and initial solid fraction $\phi_s^0 = 1 - \phi^0$,

$$\phi^{n+1} = 1 - \frac{\phi_s^0}{J^{n+1}},$$

$$\rho_{\text{mix}}^{n+1} = \phi^{n+1} \rho_f + (1 - \phi^{n+1}) \rho_s.$$

In the current code the mobility is taken as a constant scalar k .

5. Elastic Effective Stress

The total stress entering momentum balance is

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}' - p_p \mathbf{I}.$$

For `linearElastic`, the effective stress update is the incremental elastic form

$$\boldsymbol{\sigma}'_{n+1} = \boldsymbol{\sigma}'_n + \lambda \text{tr}(\Delta \boldsymbol{\varepsilon}) \mathbf{I} + 2\mu \Delta \boldsymbol{\varepsilon}, \quad \Delta \boldsymbol{\varepsilon} = \frac{1}{2} (\nabla_X \mathbf{u} + \nabla_X \mathbf{u}^T).$$

For `neoHookean`, the effective Cauchy stress is

$$\boldsymbol{\sigma}' = \frac{\mu}{J} (\mathbf{b} - \mathbf{I}) + \frac{\lambda \ln J}{J} \mathbf{I}, \quad \mathbf{b} = \mathbf{F} \mathbf{F}^T.$$

6. Particle Residuals

The particle contribution to momentum balance at test node I is

$$\mathbf{r}_{u,I}^p = (-\nabla_x N_I \cdot (\boldsymbol{\sigma}' - p_p \mathbf{I}) + N_I \rho_{\text{mix}} \mathbf{g}) v_p^{n+1}.$$

The particle contribution to mass balance is

$$r_{p,I}^p = [N_I (\ln J^{n+1} - \ln J^n) + \Delta t k \nabla_x N_I \cdot \nabla_x p_p] v_p^{n+1}.$$

The PPP stabilization residual is

$$r_{\text{PPP},I}^p = \tau_p (N_I - \bar{N}_I) [(p_p - \bar{p}_p) - (p_{p,n} - \bar{p}_{p,n})] v_p^{n+1},$$

with the same parameter choice as the reference code,

$$\tau_p = \frac{1}{2\mu}.$$

7. Variations Needed for the Hand-Written Jacobian

Take one nodal displacement component $u_{J,\alpha}$ and define

$$\delta (\nabla_X \mathbf{u}) = \mathbf{e}_\alpha \otimes \nabla_X N_J.$$

Then

$$\delta \Delta \mathbf{F} = \mathbf{e}_\alpha \otimes \nabla_X N_J, \quad \delta \mathbf{A} = -\mathbf{A} (\delta \Delta \mathbf{F}) \mathbf{A}.$$

Let

$$\mathbf{h}_\alpha = \mathbf{A} \mathbf{e}_\alpha, \quad \eta_{J\alpha} = \nabla_X N_J \cdot \mathbf{h}_\alpha.$$

Using $\nabla_x N_I = \mathbf{A} \nabla_X N_I$,

$$\delta(\nabla_x N_I) = -\mathbf{h}_\alpha (\nabla_X N_J \cdot \nabla_x N_I).$$

The Jacobian and volume variations are

$$\delta J = J \eta_{J\alpha}, \quad \delta(\ln J) = \eta_{J\alpha}, \quad \delta v = v \eta_{J\alpha}.$$

The mixture density variation is

$$\delta \rho_{\text{mix}} = (\rho_f - \rho_s) \frac{\phi_s^0}{J} \eta_{J\alpha}.$$

For the current pressure gradient

$$\mathbf{q}_p := \nabla_x p_p = \sum_K p_K \nabla_x N_K,$$

its variation is

$$\delta \mathbf{q}_p = -\mathbf{h}_\alpha (\nabla_X N_J \cdot \mathbf{q}_p).$$

8. Constitutive Tangent Actions

8.1 Linear elastic

For one column direction (J, α) ,

$$\delta \boldsymbol{\sigma}' = \lambda \text{tr}(\delta \boldsymbol{\varepsilon}) \mathbf{I} + 2\mu \delta \boldsymbol{\varepsilon},$$

with

$$\delta \boldsymbol{\varepsilon} = \frac{1}{2} (\mathbf{e}_\alpha \otimes \nabla_X N_J + \nabla_X N_J \otimes \mathbf{e}_\alpha).$$

8.2 Neo-Hookean

For

$$\boldsymbol{\sigma}' = \frac{\mu}{J} (\mathbf{b} - \mathbf{I}) + \frac{\lambda \ln J}{J} \mathbf{I}, \quad \mathbf{b} = \mathbf{F} \mathbf{F}^T,$$

the total deformation variation is

$$\delta \mathbf{F} = (\delta \Delta \mathbf{F}) \mathbf{F}^n = \mathbf{e}_\alpha \otimes \left((\mathbf{F}^n)^T \nabla_X N_J \right).$$

Then

$$\delta \mathbf{b} = \delta \mathbf{F} \mathbf{F}^T + \mathbf{F} \delta \mathbf{F}^T,$$

and

$$\delta \boldsymbol{\sigma}' = \frac{\mu}{J} \delta \mathbf{b} - \frac{\mu}{J^2} (\mathbf{b} - \mathbf{I}) \delta J + \lambda \frac{1 - \ln J}{J^2} \delta J \mathbf{I}.$$

This is the exact hand-coded `constitutive_tangent_apply()` used in the implementation; no auto-diff is used.

9. Jacobian Block Formulas

9.1 Mechanical block K_{uu}

For test node I and trial displacement (J, α) ,

$$\delta \mathbf{r}_{u,I}^p = (-\delta \boldsymbol{\sigma} \cdot \nabla_x N_I - \boldsymbol{\sigma} \cdot \delta(\nabla_x N_I) + N_I \delta \rho_{\text{mix}} \mathbf{g}) v + (-\boldsymbol{\sigma} \cdot \nabla_x N_I + N_I \rho_{\text{mix}} \mathbf{g}) \delta v.$$

This is exactly the `mech` term assembled in the code.

9.2 Coupling block K_{up}

Because only p_p depends on nodal pressure in momentum balance,

$$\frac{\partial \mathbf{r}_{u,I}^p}{\partial p_J} = v N_J \nabla_x N_I.$$

9.3 Coupling block K_{pu}

For the mass residual,

$$\delta r_{p,I}^p = N_I \delta(\ln J) v + N_I (\ln J - \ln J_n) \delta v + \Delta t k \delta(\nabla_x N_I \cdot \mathbf{q}_p) v + \Delta t k (\nabla_x N_I \cdot \mathbf{q}_p) \delta v + \delta r_{\text{PPP},I}^p.$$

Combining the first two terms gives

$$N_I (1 + \ln J - \ln J_n) \eta_{J\alpha} v.$$

For the Darcy term,

$$\delta(\nabla_x N_I \cdot \mathbf{q}_p) = -(\nabla_X N_J \cdot \nabla_x N_I) \mathbf{h}_\alpha \cdot \mathbf{q}_p - (\nabla_X N_J \cdot \mathbf{q}_p) \mathbf{h}_\alpha \cdot \nabla_x N_I.$$

Including the volume variation yields

$$\Delta t k [-(\nabla_X N_J \cdot \nabla_x N_I) \mathbf{h}_\alpha \cdot \mathbf{q}_p - (\nabla_X N_J \cdot \mathbf{q}_p) \mathbf{h}_\alpha \cdot \nabla_x N_I + (\nabla_x N_I \cdot \mathbf{q}_p) \eta_{J\alpha}] v.$$

The PPP term depends on displacement only through the current volume, hence

$$\delta r_{\text{PPP},I}^p = \tau_p (N_I - \bar{N}_I) [(p_p - \bar{p}_p) - (p_{p,n} - \bar{p}_{p,n})] \delta v.$$

9.4 Hydraulic block K_{pp}

Pressure enters mass balance through Darcy diffusion and PPP stabilization:

$$\frac{\partial r_{p,I}^p}{\partial p_J} = \Delta t k (\nabla_x N_I \cdot \nabla_x N_J) v + \tau_p (N_I - \bar{N}_I) (N_J - \bar{N}_J) v.$$

10. Verification Status

The current hand-written Jacobian has been checked against finite differences at two levels:

- particle-local verification in `examples/igampm/mpm/verify_static_twophase_local_jacobian.py`,
- multi-particle global assembly verification in `examples/igampm/mpm/verify_static_twophase_global_`

The verification covers:

- `linearElastic` and `neoHookean`,
- PPP on and PPP off,
- multiple random particle states, random shape values, random pressures, and random deformation states.

At the time of writing, the worst relative errors from these checks are:

Check	Case	Worst relative error
Local FD check	<code>linearElastic</code> , PPP off/on	$\mathcal{O}(10^{-11})$
Local FD check	<code>neoHookean</code> , PPP off/on	$\mathcal{O}(10^{-9})$
Global FD check	<code>linearElastic</code> , PPP off/on	$\mathcal{O}(10^{-11})$
Global FD check	<code>neoHookean</code> , PPP off/on	$\mathcal{O}(10^{-9})$

11. Status Table

Item	Present implementation	Status
Consistent pressure projection	yes	implemented
Updated-Lagrangian F - J kinematics	yes	implemented
Current-configuration Darcy gradient	yes	implemented
PPP stabilization with averaged shapes	yes	implemented
$K_{uu}, K_{up}, K_{pu}, K_{pp}$ hand assembly	yes	implemented
Linear elastic skeleton	yes	implemented
Neo-Hookean skeleton	yes	implemented
Auto-diff Hessian	no	intentionally not used
Plasticity from the full paper	no	pending